dopants. Four of the five valence electrons of the impurity atom form covalent bonds with four silicon atoms. The fifth valence electron of the impurity atom is loosely attached with the nucleus as it is not used in the formation of the covalent bond.

The energy level of the loosely attached fifth electron from the dopant is found just below the conduction band edge and is called the donor energy level as shown in Figure 10.5(b). At room temperature, these electrons can easily move to the conduction band with the absorption of thermal energy. It is shown in the Figure 10.6. Besides, an external electric field also can set free the loosely bound electrons and lead to conduction.

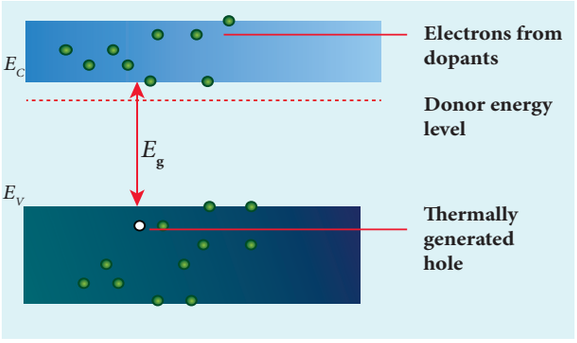


Figure 10.6 Thermally generated holes in the valence band and the free electrons generated by the dopants in the conduction band (*n*-type semiconductor)

It is important to note that the energy required for an electron to jump from the valence band to the conduction band in an intrinsic semiconductor is 0.7 eV for Ge and 1.1 eV for Si, while the energy required to set free a donor electron is only 0.01 eV for Ge and 0.05 eV for Si.

The V group pentavalent impurity atoms donate electrons to the conduction band and are called donor impurities. Therefore, each impurity atom provides one extra electron to the conduction band in addition to the thermally generated electrons. These thermally generated electrons leave holes in valence band. Hence, the majority carriers of current in an *n*-type semiconductor are electrons and the minority carriers are holes. Such a semiconductor doped with a pentavalent impurity is called an *n*-type semiconductor.

ii) *p*-type semiconductor

In *p*-type semiconductor, trivalent impurity atoms (from III group of periodic table) such as boron, aluminium, gallium and indium are added to the silicon (or germanium) crystal. The dopant with three valence electrons can form three covalent bonds with three silicon atoms. Of the four covalent bonds, three bonds are complete and the remaining one bond is incomplete with one electron. This electron vacancy present in the fourth covalent bond is represented as a hole.

To make complete covalent bonding with all four neighbouring atoms, the dopant is in need of one more electron. These dopants can accept electrons from the neighbouring atoms. Therefore, this impurity is called an acceptor impurity. The energy level of the hole created by each impurity atom is just above the valence band and is called the acceptor energy level, as shown in Figure 10.7(b).

For each acceptor atom, there will be a hole in the valence band; this is in addition to the holes left by the thermally generated electrons. In such an extrinsic semiconductor, holes are the majority carriers and thermally generated electrons are minority carriers as shown in Figure 10.8. The extrinsic semiconductor thus formed is called a *p*-type semiconductor.

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